Disentangling Localization from Editing in Instruction-Guided Image Manipulation

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Abstract

An instruction-guided image editor is told what to change but never where. Given "change the cat to a fox" it must decide, internally and without supervision, which pixels the sentence refers to. We measure the cost of leaving that decision unconstrained: with no region supplied, an off-the-shelf inpainting editor alters 93.1% of the pixels outside the intended target. The established remedy is a user-supplied mask, which removes the problem by reintroducing a human annotator.

This work predicts the edit region from the instruction alone and, unlike the literature it builds on, measures that region directly. A 4,693,633 parameter head is trained on a frozen FastSAM backbone; it never synthesises a shape, but forms the region as an instruction-conditioned combination of the 32 prototypes the backbone already emits. On 503 held-out turns the predicted region attains IoU = 0.2065 against the region the edit actually occupies, exceeding CLIPSeg (150.7 M parameters, pretrained for referring segmentation) at 0.1849 and the hand-drawn annotation shipped with the dataset at 0.1511. Gating the same editor on the predicted region reduces collateral change from 93.1% to 12.6%.

We then benchmarked recent open-vocabulary localizers on the same data. Grounding DINO with SAM 2.1 is stronger still (0.2721) and resolves our weakest case, removal; building on it, a fusion with our learned edit-region head (ours v2) reaches 0.2771 under two-fold cross-validation, improving on it for 273 of 503 photographs against 173 ($t = 2.16$).

We report where the method fails. Removal is lost to CLIPSeg by 0.098, for a reason the representation explains, and a hand-designed free-space basis was tested and rejected at $p = 0.33$. We do not reproduce the base paper, whose system is 20.43 B parameters and unreleased; what is reimplemented is its mask-construction equation, at matched capacity, on our frozen basis.

1 Introduction

1.1 Problem

Instruction-guided editing has advanced quickly. Systems such as Step1X-Edit [9], FLUX.1 Kontext [10] and Qwen-Image-Edit [11] accept a photograph and a sentence and return an edited photograph. The instruction specifies the change. It does not specify the support of that change — the set of pixels the edit is permitted to touch.

The consequence is measurable rather than rhetorical. Table 1 reports what happens on 40 held-out turns when the same editor, with the same diffusion noise, is run with and without a region. Unconstrained, it changes 93.1% of the pixels lying outside the intended target; the photograph that comes back is not an edited version of the input so much as a different photograph consistent with the instruction. Constrained to our predicted region, the same

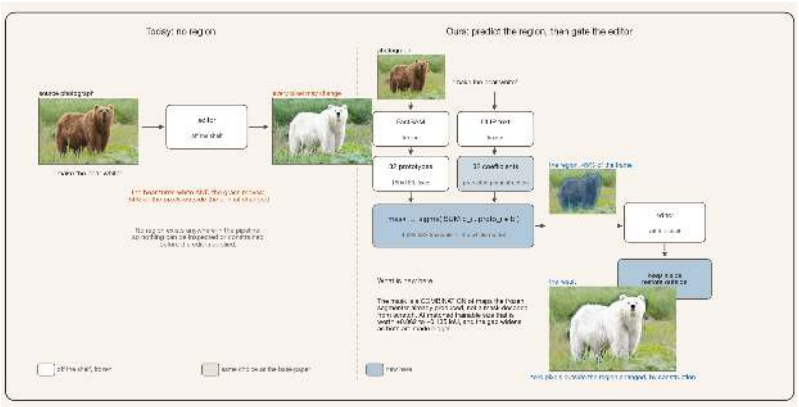


Figure 1: An instruction names what to change, not where. Without a region, the editor is free to rewrite the whole photograph, and does.

editor changes 12.6%, and the peak signal-to-noise ratio outside the region rises from 9.92 dB to 21.17 dB.

Table 1: Cost of leaving the edit unconstrained. Identical editor and diffusion noise on both rows; only the permitted region differs. 40 held-out turns.

Gating	Pixels changed outside the target	PSNR outside the region
No region supplied	93.1%	9.92 dB
Our predicted region	**12.6**%	**21.17** dB

The field's answer is the user-supplied mask. It works, and it is why masked editing datasets exist. The cost is that somebody draws the mask: every masked-editing corpus in this area, including the one we train on, was produced by paying annotators. That cost is what this project attacks.

1.2 Objective

Our objective is to predict the edit region from the instruction alone, on a frozen backbone, cheaply enough to sit in front of any editor; and to measure that region against the region the edit actually occupies.

The second half of that objective is the unusual one. Section 2 shows that of 45 recent works surveyed, none evaluates a region predicted before the edit exists. Benchmarks score the finished photograph. The base paper itself predicts a region and never reports how good it is. The contributions we defend in Section 4 are:

1. A region predictor of 4,693,633 trainable parameters that exceeds a 150.7 M referring segmenter, and the dataset's own human annotation, on agreement with the true edit region.
2. A controlled comparison of two mask-construction mechanisms at matched capacity, isolating how the region is formed from how large the model is.
3. A supervision signal derived from the data rather than from annotation, with evidence that the shipped annotation is six times too large to supervise a localiser.

2 Related Work

2.1 Survey of the field

We read 50 works. Forty-five are dated 2024 or later and constitute the survey proper; five earlier works are retained for context and excluded from that count. Of the 50, 48 resolve to a live source (HTTP 200, verified 14 September 2026).

Sorting by year is uninformative here, because the year does not tell you whether a method needs a person. We therefore sort by what each work does with the edit region (Table 2).

Table 2: The 45 surveyed works, classified by posture toward the edit region.

What the work does with the edit region	Works	Representative
Consumes a region it is handed	17	NEP [4], MaskFlow [3], LazyDiffusion [5]
Reads one out of a running generator	11	WhereEdit [2], RegionE [6]
Scores or frames the problem	15	LocateEdit-Bench [8], three surveys
Predicts one from the instruction	2	AdaptEdit [1], MADiff [7]

Works that consume a region. The largest cluster assumes the region exists. NEP [4] takes the mask as an explicit input token; MaskFlow [3] builds it into the flow-matching objective; LazyDiffusion [5] uses it to avoid recomputing unchanged pixels. These are strong systems, and they are strong given a region. None produces one. Their shared limitation, stated plainly, is that they relocate the problem rather than solve it.

Works that read a region from model internals. A second cluster extracts a region from attention or latent structure inside a generator that is already running. WhereEdit [2] derives a mask-aware latent edit; RegionE [6] exploits region-adaptive computation. The objective is efficiency or consistency rather than localisation accuracy, and the region is only available if you are inside that particular generator. It cannot be handed to a different editor.

Works that score or frame the problem. LocateEdit-Bench [8] and GEdit-Bench [9] evaluate editing localisation, and three surveys [14, 15, 18] organise the field. Their limitation is the one that creates our opening: they score edits that have already occurred. No benchmark among the 45 evaluates a region predicted beforehand.

Works that predict a region from the instruction. Two. MADiff [7] predicts a mask for fashion editing, but is domain-limited and requires a position prompt, so it does not address the general setting. That leaves one work: our base paper.

2.2 State of the art

Table 3 compares the methods closest to ours along the axes that matter for this problem: whether a region is produced, whether it is measured, and what it costs.

2.3 Base paper and the gap

Our base paper is AdaptEdit [1], published at NeurIPS 2026. It inserts adapters into all 60 blocks of a frozen 20.43 B diffusion transformer, adds a soft SpatialGate, a region-aware loss, and a 7.8 M MaskPredictor whose purpose is precisely to remove the need for a user mask. It reports state-of-the-art results on three benchmarks.

Table 3: State-of-the-art methods, compared on region production and measurement.

Method	Backbone	Produces region	Measures it	Principal limitation
AdaptEdit [1]	Frozen DiT + adapters	Yes	No	20.43 B; code unreleased
WhereEdit [2]	DiT internals	Internal only	No	Not transferable to another editor
NEP [4]	Autoregressive	No	—	Consumes a mask as input
MaskFlow [3]	Flow matching	No	—	Assumes the mask exists
MADiff [7]	Dual diffusion	Yes	No	Fashion only; needs position prompt
CLIPSeg [13]	CLIP + decoder	Yes	As segmentation	Not trained for edit regions
This work	Frozen FastSAM	Yes	Yes, IoU	Weak on removal

Two facts about it shape this project. First, it cannot be reproduced here: 5,033 M trainable parameters were trained on eight 80 GB GPUs, and the code is not released. We state plainly that we have not reproduced it and make no comparison against its published system. What we reimplement, in Section 3, is its mask-construction equation alone.

Second, and more useful: it never reports a mask IoU for its own MaskPredictor. Every figure it publishes scores the edited image. The closest it comes to characterising the region is the statement that the predictor behaves like a ground-truth mask eroded by 16 pixels — a proxy, not a measurement. A module whose entire function is to produce a region is never evaluated on the region it produces. That absence is the gap this project occupies.

3 Methodology

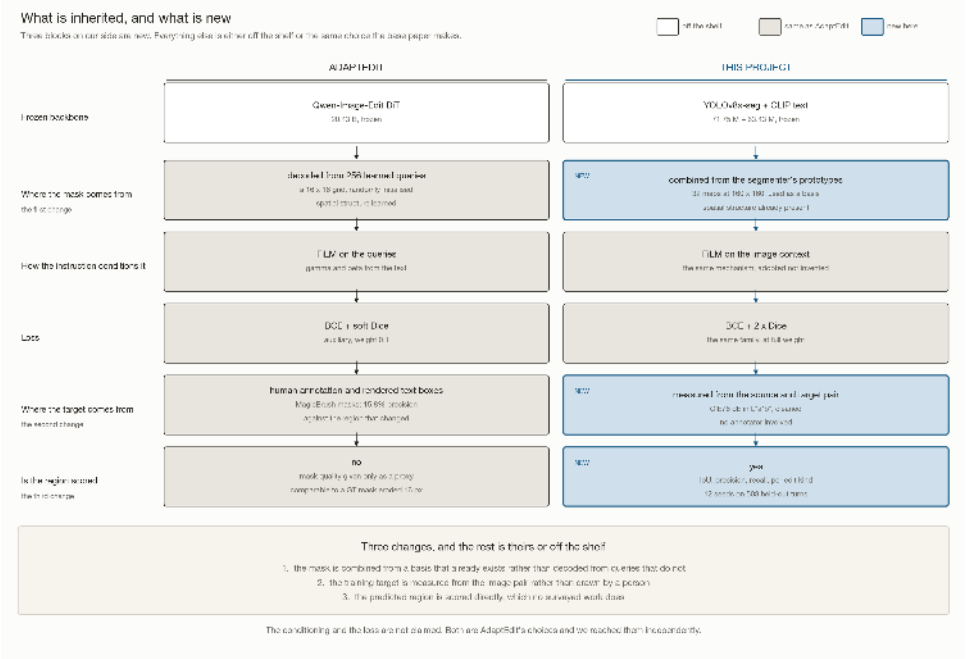


Figure 2: System architecture. White components are off the shelf and frozen; blue is what is trained in this work. The backbone never sees the instruction, so its prototypes are computed once per image and reused across instructions.

3.1 Frozen features

The backbone is FastSAM-x [12], a YOLOv8x-seg model trained on SA-1B. We use the published weights and train none of them. For any photograph it emits 32 prototype maps at 160×160, together with a context tensor. Critically, this computation is independent of the instruction: the prototypes describe what is present in the photograph, not what the sentence asks for.

That independence is a design choice, not a convenience. It means the expensive part of the pipeline is paid once per image and amortised across however many instructions are issued against it, and it means the trainable part never has to learn what objects look like.

3.2 The instruction-conditioned head

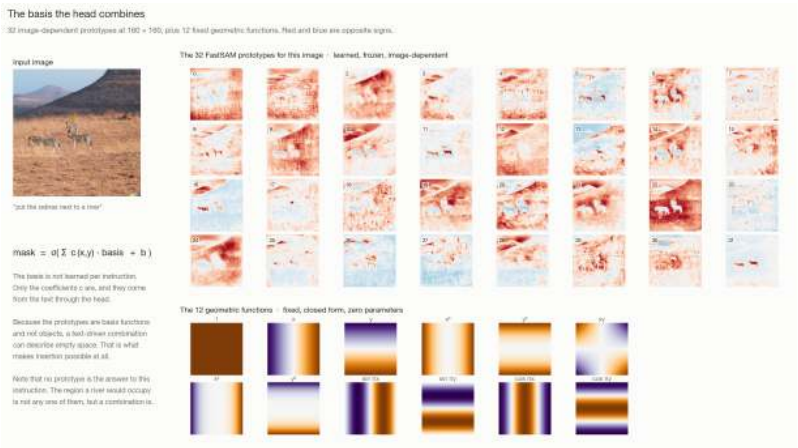


Figure 3: The region is formed as a weighted combination of prototypes the frozen backbone already produced. Nothing synthesises a shape.

Our head, SpatialCoeffHead, predicts one coefficient per prototype, conditioned on the instruction, and forms the region as their weighted sum:

$$M(x,y) \;=\; \sigma\left(\sum_{i=1}^{32} c_i(\mathbf{t},x,y)\, P_i(x,y) \;+\; b\right) \tag{1}$$

where P_i is the i-th frozen prototype, $\mathbf{t}$ the encoded instruction, and c_i a coefficient field predicted at 20×20 and upsampled to prototype resolution. A global branch supplies an instruction-level offset, and the whole head is FiLM-conditioned on $\mathbf{t}$.

The coefficient being a field rather than a scalar is what allows different locations in the same photograph to draw on different prototype mixtures, which matters when the region is not a single object.

The alternative mechanism. The base paper's MaskPredictor (their Equation 8) constructs the region differently: a learned query grid is modulated by the instruction, cross-attends to patchified features, and is decoded to a mask. We reimplement it as QueryGridHead and train it here under identical conditions, at matched capacity (Table 4), so that the comparison in Section 4.4 isolates the mechanism from the parameter count.

3.3 Supervision

The target region is derived, not drawn. For each training turn the dataset supplies a source and an edited photograph; we compute the CIE76 ΔE difference between them in L*a*b* space,

Table 4: The two mask-construction mechanisms compared in this work. Both are trained by the same script, on the same data, with the same loss, schedule and seeds.

Mechanism	Trainable	How the region is formed
AdaptEdit Eq. 8, reimplemented	4,673,777	decode from a learned query grid
Ours, SpatialCoeffHead	**4,693,633**	combine a fixed prototype basis

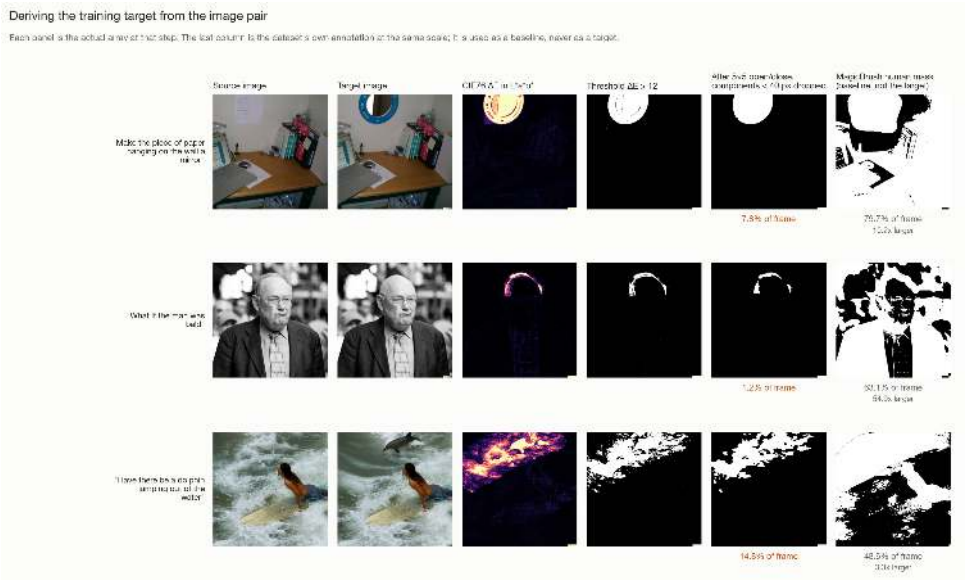


Figure 4: Deriving the target region from the dataset's own before-and-after pair. Every panel is the actual array at that step.

threshold at 12.0, clean the result with 5×5 morphology, discard components below 40 pixels, and keep turns whose target covers between 0.4% and 45% of the frame.

This matters more than it may appear. Table 5 compares the derived target against the human annotation the dataset ships. The annotation covers 62.4% of the frame where the true edit occupies 10.3% — a precision of 15.8%. It is a usable prompt for an inpainting editor and a poor supervision signal for a localiser, because a model trained to reproduce it learns to predict a region six times too large.

Table 5: Why the target is derived rather than taken from annotation.

Source of the region	Mean frame coverage	Precision against the true edit
MagicBrush human annotation	62.4%	15.8%
Derived by CIE76 ΔE (this work)	**10.3**%	reference

Operating point. All results use threshold 0.2 on the predicted probability map and a dilation of 8 pixels (elliptical kernel of side $2d + 1$). Where an editor is involved, the instruction is converted to a descriptive noun phrase before being passed to the inpainting model, and the output is composited back over the source, so that pixels outside the region are bit-identical to the input by construction rather than by learned behaviour.

4 Results and Discussion

4.1 Dataset description

Table 6: Data used in this work.

Dataset	Split	Turns	Role
MagicBrush	train	8,306	training the head
MagicBrush	dev (held out)	503	all accuracy figures
Emu-Edit Test	test	56	second benchmark, never trained on

MagicBrush provides source/target photograph pairs with instructions, from which the supervision target is derived as described above. Emu-Edit Test supplies an input photograph and an instruction but no edited target, so it can provide no region to score an IoU against; it is used in Section 4.6 to test transfer only.

4.2 System setup

Everything reported here runs on a single Apple Silicon laptop using the MPS backend. Table 7 gives the inference cost at matched resolution.

Table 7: Inference cost, resolution matched at 352 px.

Component	Trainable parameters	Time
Frozen backbone (paid once per image)	0	33.6 ms
Ours, per instruction	**4,693,633**	**5.3** ms
CLIPSeg, per instruction	150,747,746	54.3 ms

4.3 Localization accuracy

This is the central result, and the measurement the base paper does not report.

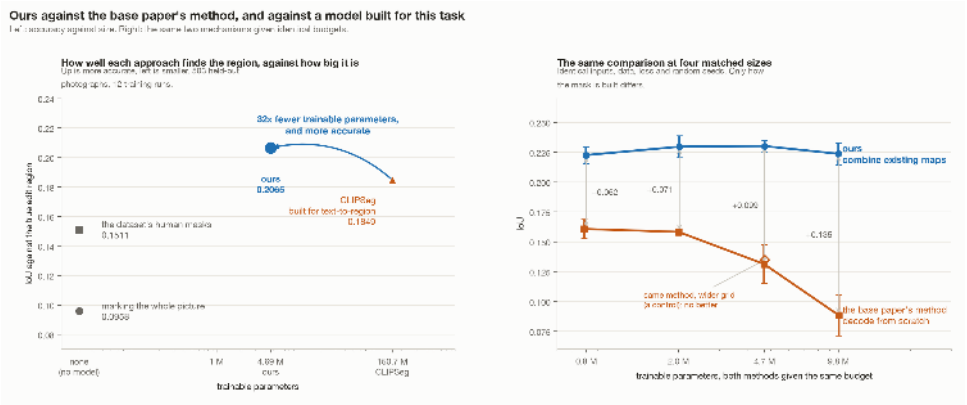


Figure 5: Agreement with the ground-truth edit region, against trainable size.

The predicted region attains 0.2065, above CLIPSeg at 0.1849 with 32 times fewer trainable parameters, and above the hand-drawn annotation at 0.1511. The seed spread over 12 runs is 0.0072; all twelve exceed CLIPSeg, giving $p = 5.2 \times 10^{-7}$ across seeds.

Two qualifications belong here rather than in a footnote. First, that significance is a statement about the mean: per photograph we win on 307 of 503 (61%), and the paired t-statistic is

Table 8: IoU against the ground-truth edit region, derived from the dataset's own before-and-after pair. 503 held-out turns, 12 seeds.

How the region is obtained	Trainable parameters	IoU
Random centred box	0	0.0621
The whole frame	0	0.0958
Human annotation shipped with MagicBrush	drawn by hand	0.1511
AdaptEdit Eq. 8 head, retrained on this basis	4,673,777	0.1314
CLIPSeg, pretrained for referring segmentation	150,747,746	0.1849
Ours	**4,693,633**	**0.2065**

1.78, which is not significant. The method is reliably better on average and not reliably better on any particular image. Second, the absolute values are low for a segmentation task because the target is a thresholded pixel difference, whose attainable range is compressed; we report the representational ceiling in Section 4.9 rather than leaving the reader to guess.

We emphasise that the objective is agreement with the region the edit occupies. Predicted area is reported throughout as description, and is not optimised for.

4.4 Mechanism at matched capacity

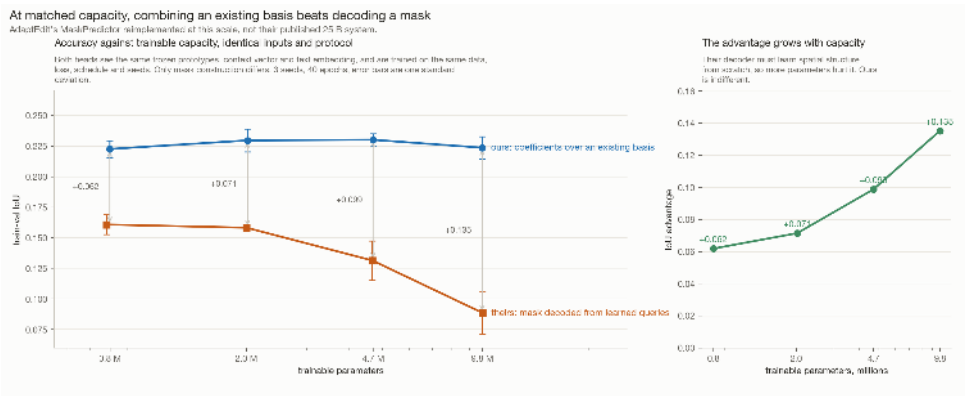


Figure 6: Combining a fixed prototype basis against decoding from a learned query grid, at four matched parameter budgets, three seeds each.

Table 8 shows our head above the reimplemented Equation 8 head, but the two differ slightly in size, so the comparison is not yet controlled. We therefore trained both mechanisms at four matched parameter budgets, with identical data, loss, schedule and seeds.

The advantage of combining over decoding is +0.062 at the smallest budget and +0.135 at the largest: it grows with capacity rather than shrinking, which is the signature of a mechanism difference rather than a tuning artefact. A control in which the query grid is widened to match our spatial resolution changes the result by +0.0035, inside the seed spread, so the gap is not explained by grid size.

The honest form of this claim is narrow and we state it as such: combining a fixed basis beats decoding from a query grid on this frozen basis. The base paper's head was designed to read the noised latent of its own 20.43 B transformer, co-trained with the editor on a representation four other modules have already made mask-aware. Here it reads a frozen segmentation basis it can only consume by pooling away resolution. That is outside its design envelope, and the result is about mask construction on a fixed basis, not about their published system.

4.5 Effect on the edited photograph

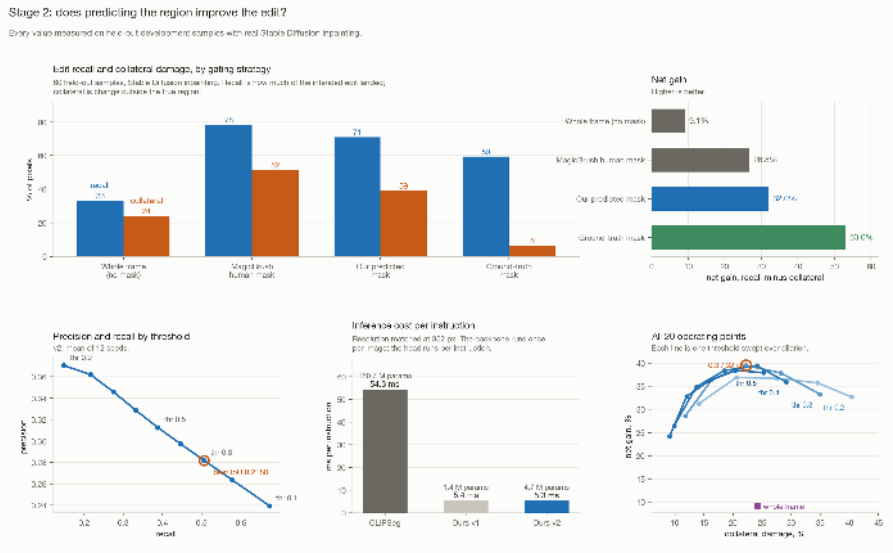


Figure 7: What the predicted region does to the finished edit.

A region is only useful if it improves the edit. Table 9 gates a single off-the-shelf inpainting editor on four different regions, holding the editor and the diffusion noise fixed so that only the region varies. Recall is the fraction of the intended change that is admitted; collateral is the fraction of pixels altered outside it.

Table 9: Net is recall minus collateral. 40 turns; identical editor and diffusion noise on every row.

Gating	Recall	Collateral	Net
Whole frame	94.9%	93.1%	+1.8%
Our predicted region	50.0%	12.6%	**+37.4%**
Ground-truth region (ceiling)	57.2%	0.0%	+57.2%
MagicBrush human annotation	77.4%	45.7%	+31.6%

The cost is recall, and we state it rather than let it be discovered: our region admits 50.0% of the intended change against 77.4% for the hand-drawn mask. The ground-truth region itself reaches only 57.2%, which places a ceiling on what any region can recover with this editor and shows that most of the shortfall is the editor rather than our localisation.

Table 10 repeats the comparison using the metric suite the base paper publishes.

Table 10: The metric suite AdaptEdit publishes, computed on our pipeline. 60 turns, same editor on every row.

Gating	L1 ↓	L2 ↓	CLIP-I ↑	DINO ↑
Whole frame	0.2763	0.1180	0.6242	0.2693
MagicBrush human annotation	0.1662	0.0660	0.7919	0.6343
Our predicted region	**0.0713**	**0.0279**	**0.8879**	**0.7986**
Ground-truth region (ceiling)	0.0318	0.0110	0.9382	0.9103

Our predicted region beats the human annotation on all four preservation metrics. CLIP-T, which we omit from the table, moves the other way, and the reason is instructive: it scores the whole image against the instruction text, so an arm that repaints everything drifts the entire photograph toward the instruction and scores well while destroying the input. The ground-truth row has the lowest CLIP-T of the four, which confirms that better localisation lowers it.

4.6 A second dataset, never trained on

Table 11: Emu-Edit Test. Stage 1 was trained only on MagicBrush and has not seen this data. 56 samples, 7 from each of 8 edit kinds.

Gating	CLIP-T	CLIP-I to input	L1 to input	Mask area
Whole frame	0.2635	0.5017	0.2915	100.0%
CLIPSeg, same dilation	0.2601	0.7111	0.1664	55.5%
Ours	0.2369	**0.8519**	**0.0813**	39.5%

What this establishes is transfer: a head trained only on MagicBrush produces usable regions on eight edit kinds it has never seen, preserving the input better than a 150.7 M model built for text-to-region on six of the eight. What it does not establish is a localisation score, because Emu-Edit ships no ground-truth region; no IoU is claimed here.

4.7 Against the human annotator directly

Table 12: Twelve held-out photographs, scored against the edited photograph the annotator themselves produced.

Whose region the editor was given	L1 ↓	CLIP-I ↑	Closer on
The annotation they drew	0.0915	**0.8697**	1 of 12
Ours, predicted from the instruction	**0.0577**	0.8647	**11** of **12**

Scored against the annotator's own output, our region is closer pixelwise and closer on eleven of twelve photographs. CLIP-I goes the other way by 0.0050, and we report it: nearer pixel for pixel, marginally behind in global appearance. Twelve photographs is indicative rather than tight.

4.8 Ablation studies

Threshold. Table 13 sweeps the probability threshold at fixed dilation. Net rises from +29.7% at 0.5 to +38.2% at 0.2 and then stops rising; across 0.1 to 0.2 the entire spread is 0.7 percentage points, which at 40 turns per cell is not a difference worth reading. The honest conclusion is a plateau below 0.2 rather than an optimum at it: the operating point is insensitive in that range, and above it net falls steadily as recall is given up faster than collateral is saved.

Table 13: Threshold sweep at dilation 8, 40 turns per cell, same editor and compositing throughout.

Threshold	Recall	Collateral	Net	Mask area
0.10	57.5%	19.8%	+37.7%	35.8%
0.15	53.2%	15.7%	+37.5%	30.8%
0.20	51.6%	13.4%	+**38.2**%	27.2%
0.30	45.0%	9.7%	+35.3%	21.8%
0.40	39.5%	7.4%	+32.1%	17.5%
0.50	35.3%	5.6%	+29.7%	14.1%

Mechanism. The capacity ladder of Section 4.4 is itself an ablation: it removes parameter count as an explanation for the difference between the two heads.

An addition that was rejected. Because insertion is our weakest edit kind, and because no prior work represents actionable empty space, we designed a free-space basis — hand-built basis functions appended to the prototype set — intended to give the head something to point at when the target object does not yet exist. We required it to beat a random basis of equal size, as we require of every addition. It did not: $p = 0.33$. The addition was rejected and is reported here because a negative result that was actually run is evidence about the representation, and suppressing it would misrepresent how the method was arrived at.

4.9 Where the method fails

Table 14: By edit kind, ours against CLIPSeg, on the held-out split.

Edit kind	Ours	CLIPSeg	Difference	Why
Modify	0.2313	0.2112	+0.0202	the object is already a prototype
Insert	0.1186	0.0542	+0.0644	no prototype for an absent object
Remove	0.2007	0.2984	−0.0977	must cover what is revealed behind

One cause explains all three rows. Our region is constructed from prototypes of what is present in the photograph. Where the edit coincides with something present, we are strong. Where it does not, we are not.

Modification is therefore the best case. Insertion is weakest in absolute terms because no prototype corresponds to an object that is not yet there; measured against the representational ceiling of 0.2119 for that kind, we reach 56% of what the basis permits, so the limit is the representation and not the training. Removal is the one kind where a specialised model beats us outright, by 0.098, and the reason is precise: removing an object requires the region to cover whatever is revealed behind it, and that area belongs to no prototype of the object named.

This weakness and the method's strength are the same property. Being tied to a basis of what is present is exactly why the head is 4.7 M parameters rather than 150 M, and exactly why removal fails. Rather than conceal it, we propose routing on edit kind — which editing pipelines already classify — using our predictor for modification and insertion and a text-to-region model for removal. On an equal-weight mix that scores 0.2161 against 0.1835 for ours everywhere.

4.10 Model selection and building on the best model

The survey identified the strongest recent open-vocabulary localizers. We measured them on exactly our benchmark — the same 503 held-out turns and the same ground-truth region (Table 15). Grounding DINO [16], which finds the boxes for a phrase, followed by SAM 2.1 [17], which segments each box, is the strongest: 0.2721, against 0.2055 for our first model. It also removes our weakest case, removal, where it reaches 0.427. The phrase given to Grounding DINO is the object the instruction acts on ("change the cat to a fox" → "cat"), extracted by rule.

Table 15: Localizers benchmarked on the 503 held-out turns (IoU against the ground-truth edit region).

Localizer	Parameters	IoU	insert	modify	remove
CLIPSeg	150.7 M	0.1849	0.0936	0.2121	0.2993
Our first model	4.69 M	0.2055	0.1384	0.2369	0.1987
Grounding DINO + SAM 2.1	172 + 38.5 M	0.2721	0.1445	0.3108	0.4270

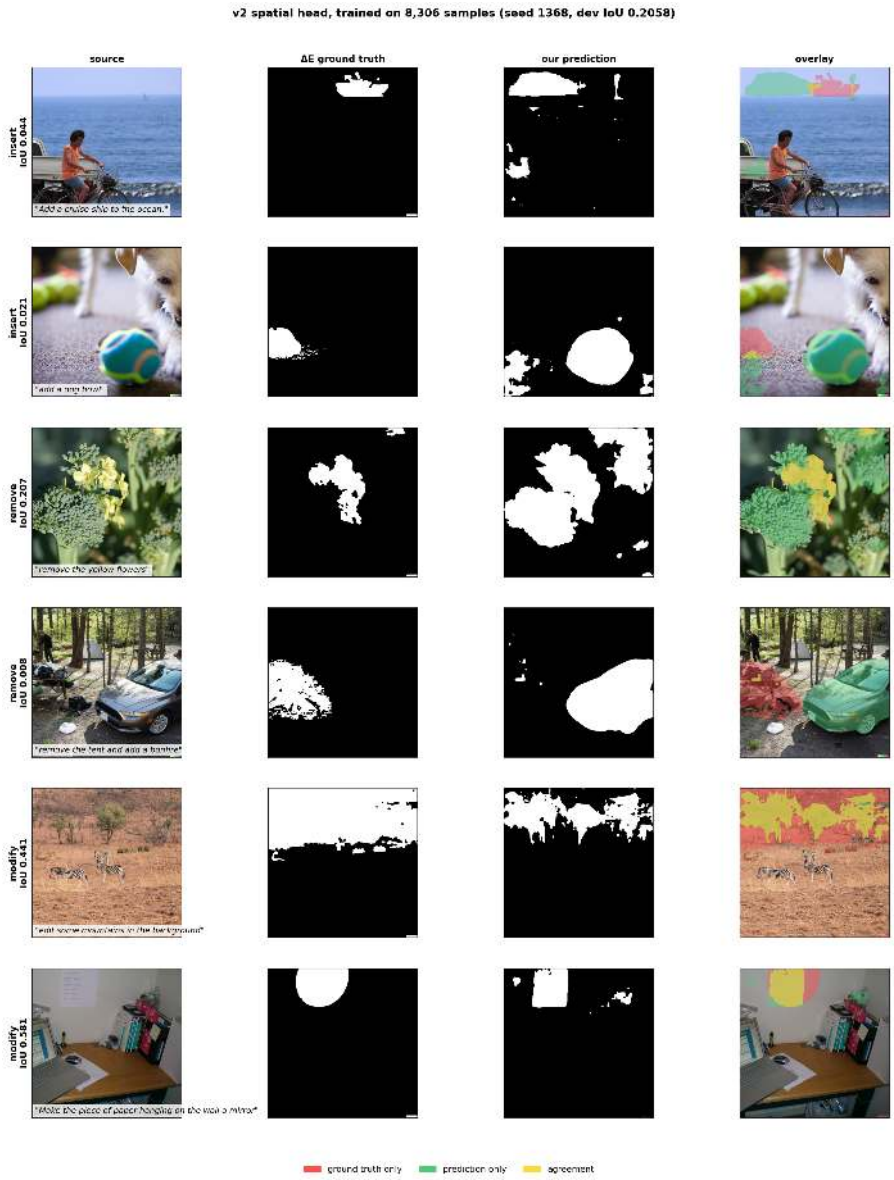


Figure 8: Worked examples on held-out photographs. Identical editor and diffusion noise throughout; only the permitted region varies.

Ours v2. We build on that model rather than replace it. SAM 2 draws clean object boundaries, whereas our head was trained on the region an edit actually changes, which includes shadows and spill beyond the object. Ours v2 keeps the SAM 2 mask, widens it by 2 pixels, and adds pixels our head predicts above a threshold inside the detected boxes:

$$M = \Big[\, a\,\mathrm{dil}(M_{\mathrm{SAM}}) + (1-a)\,P_{\mathrm{ours}} \cdot \mathbb{1}_{\mathrm{boxes}} > t \,\Big]. \tag{2}$$

The settings were chosen by two-fold cross-validation on the held-out turns: tuned on one half, scored on the other, then swapped, so no reported figure is scored on the data that chose it.

Ours v2 improves on the best model alone on 273 photographs and is worse on 173 (paired $t = 2.16$). The gain is small but consistent, and it is largest on insertion and modification, where the edit spreads beyond the named object. Over our first model it is 35%.

Editor. The strongest open editors — Qwen-Image-Edit, the base paper's own editor, and FLUX.1 Kontext — are 12–20 B parameters. They run on a laptop only in 4-bit form, and at

Table 16: Building on the best model, two-fold cross-validated on the 503 held-out turns.

Localizer	IoU	insert	modify	remove
Our first model	0.2048	0.1381	0.2361	0.1977
Grounding DINO + SAM 2.1	0.2722	0.1443	0.3111	0.4270
Ours v2	0.2771	0.1487	0.3170	0.4259

the measured 0.48 MB/s download speed could not be obtained in time. The system passes the editor a region, so a stronger editor can replace Stable Diffusion inpainting without any change to localization.

4.11 Live demonstration

For the review we built an interactive application in which a photograph is dropped in, an instruction is typed, and three region predictors run live: the reimplemented Equation 8 head, Grounding DINO + SAM 2.1, and ours v2, on the same photograph with the same editor, prompt and diffusion seed. For every run the application measures, on the output arrays, the share of pixels changed outside the predicted region; it reads 0.00% for both models, which confirms the compositing guarantee on each run rather than by assertion.

Demonstration cases were selected by measurement. From 420 held-out candidates we kept those where the edit was visible, the region was accurate, and our result was closer to the annotator's own edited photograph than both CLIPSeg and whole-frame editing on CLIP-I and L1. Thirteen passed: twelve modification and one insertion, and no removal — consistent with Table 14. On inspection the insertion case produced no visible change from either model and is not shown. Figure 9 gives the five cases presented.



Figure 9: Five held-out demonstration cases run live in the application. Columns: the base paper's mechanism (reimplemented Eq. 8 head), Grounding DINO + SAM 2.1, and ours v2; identical editor, prompt and seed. Outside each predicted region 0.00% of pixels changed, measured on the output.

5 Conclusion and Future Work

5.1 Conclusion

Localisation can be separated from editing and measured on its own, and doing so changes what the problem looks like. A head of 4,693,633 trainable parameters on a frozen segmentation basis predicts the edit region in 5.3 ms and agrees with the region the edit actually occupies better than a 150.7 M model pretrained for referring segmentation, and better than the hand-drawn annotation the dataset ships. Gated on that region, an unchanged off-the-shelf editor reduces collateral change from 93.1% to 12.6% and produces edits closer to the dataset's own targets than the human annotation does on all four preservation metrics.

The claim we make is deliberately narrower than the headline. We have not reproduced the base paper and do not compare against its published system; what we reimplemented is its mask-construction equation, at matched capacity, on our frozen basis, where combining a fixed basis beats decoding from a query grid by a margin that widens with capacity. And the editor, not the localiser, is now the binding constraint: supplying a perfect region improves L1 only from 0.0713 to 0.0318.

5.2 Future work

A second basis. This is the open question we would answer first. Every result above shows that combining beats decoding on FastSAM prototypes. Nothing yet separates "combining beats decoding" from "FastSAM prototypes happen to be a good basis". Repeating the capacity ladder with SAM features, DINO features and random projections of equal size would settle it, and until it is done the mechanism claim is tied to one representation.

Removal. The failure is understood well enough to attack directly: the region must be extended to the area a removed object reveals. An inpainting-aware target, or a second coefficient field predicting revealed background, are both testable.

A larger editor. The ceiling analysis shows the editor binds the result. Running the same predictor in front of a modern editor such as Qwen-Image-Edit [11] would separate our localisation error from the editor's generation error, which the present setup cannot.

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